INCH-POUND
MIL-PRF-39012/57F
27 January 1992
SUPERSEDING
MIL-C-39012/57E
3 October 1986

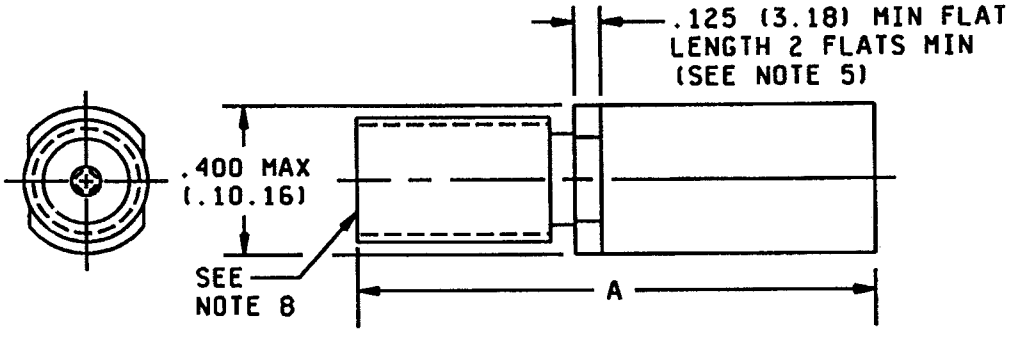
NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

PERFORMANCE SPECIFICATION

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY, (SERIES SMA (CABLED) - SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-39012.



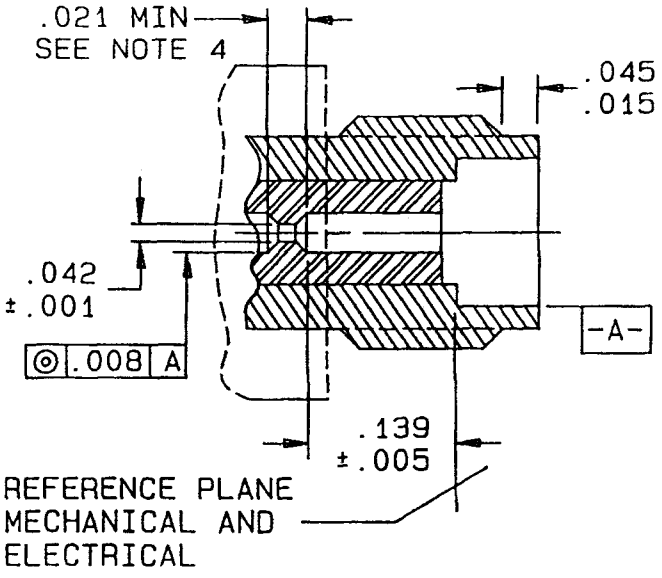

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE VII

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see tables I and V.
4. Dimension .400 (10.16) is the largest overall diameter of the connector.
5. Wrench flats are to accommodate standard wrench openings in accordance with FED-STD-H28, appendix 10.
6. Dimension A defines the overall length of connector when assembled to the cable.
7. All undimensioned pictoral configurations are for reference purposes only.
8. Series SMA, socket contact interface in accordance with MIL-STD-348.
9. Metric equivalents are in parentheses.

FIGURE 1. General configuration.


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CATEGORY D

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .001 | 0.03 | .0355 | 0.902 | .115 | 2.92 |
| .002 | 0.05 | .0370 | 0.940 | .139 | 3.53 |
| .005 | 0.13 | .042 | 1.07 | .168 | 4.27 |
| .008 | 0.20 | .045 | 1.14 | .170 | 4.32 |
| .010 | 0.25 | .049 | 1.24 | .1810 | 4.60 |
| .015 | 0.38 | .051 | 1.30 | .208 | 5.28 |
| .021 | 0.53 | .074 | 1.88 | .216 | 5.49 |
| .030 | 0.76 | .078 | 1.98 | .218 | 5.54 |
| | | | | .250 | 6.35 |

NOTES:
1. Dimensions are in inches.
2. Slitting of inner contact optional.
3. Metric equivalents are given for general information only.
4. Chamfer is optional, if chamfer is used put on a 30° maximum.

FIGURE 2. Category D captivation detail.



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TABLE I. Dash numbers, cross reference and dimensions.

| Dash number | Applicable cable ¤ | Dimensions | Inches-millimeters maximum 1/ |
|---|---|---|---|
| Category A - Field serviceable (no special tools required) 2/ | | | |
| 3006 3/<br>4006 3/ | M17/93-RG178*<br>M17/169-00001ϕ | A | .960(24.38) |
| 3007 3/<br>4007 3/ | M17/119-RG174<br>M17/173-00001ϕ<br>M17/113-RG316*<br>M17/172-00001ϕ | | |
| 3008 3/<br>4008 3/ | M17/54-RG122*<br>M17/157-00001ϕ | | |
| 3009 3/<br>4009 3/ | M17/28-RG058<br>M17/60-RG142@<br>M17/84-RG223*<br>M17/155-00001ϕ<br>M17/158-00001ϕ<br>M17/167-00001ϕ | | |
| 3010 3/<br>4010 3/ | M17/111-RG303*@<br>M17/170-00001ϕ | | |
| 3030 3/<br>4030 3/ | M17/152-00001 | | |
| Category C - Field replaceable (MIL-C-22520/5-01 basic crimp tool) 4/ | | | |
| 3025 3/<br>4025 3/ | M17/93-RG178∞*<br>M17/169-00001∞ϕ | A | 1.265(32.13) |
| 3026 3/<br>4026 3/ | M17/119-RG174Σ<br>M17/173-00001Σϕ<br>M17/113-RG316Σ* | | |
| 3027 3/<br>4027 3/ | M17/54-RG122*&<br>M17/157-00001ϕ& | | |

See footnotes at end of table.



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TABLE I. Dash numbers, cross reference and dimensions - Continued.

| Dash number | Applicable cable ¤ | Dimensions | Inches-millimeters maximum 1/ |
|---|---|---|---|
| Category C - Field replaceable (MIL-C-22520 crimp tool) 4/ | | | |
| 3028 3/<br>4028 3/ | M17/60-RG142@~*<br>M17/158-00001-ϕ<br>M17/167-00001-ϕ<br>M17/84-RG223≈ | A | 1.265(32.13) |
| 3029 3/<br>4029 3/ | M17/155-00001-ϕ<br>M17/28-RG058*~<br>M17/111-RG303~<br>M17/170-00001-ϕ | | |
| Category D - Field replaceable - Defined piece parts 4/ 6/ 5/ | | | |
| [illegible] 3/<br>4502 3/ | M17/60-RG142@*<br>M17/158-00001ϕ<br>M17/128-RG400<br>M17/175-00001ϕ | A | 1.265(32.13) |

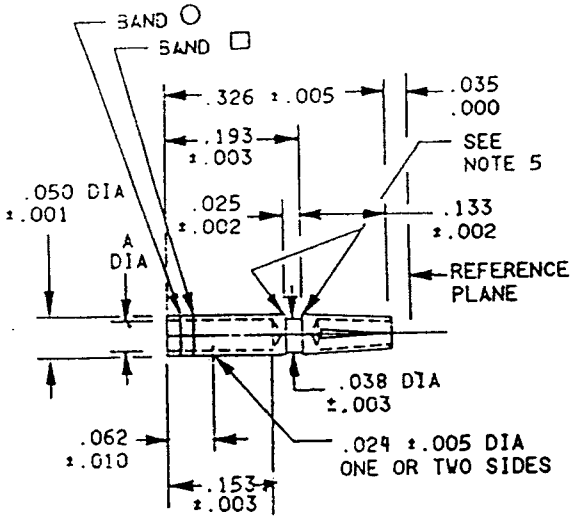
1/ Millimeters are in parentheses.
2/ All corrosion resistant steel bodied connectors which are designed to be assembled to the cable outer conductor using solder shall be gold plated in accordance with MIL-G-45204, type II, class I.
3/ These connectors have captivated center contacts.
4/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 4.
5/ Not for use in army equipment.
6/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

¤ The latest version of each cable shall be applicable.
★ Cable to be used when performing tests requiring cable except as in note Σ.
@ Cable to be used for the +200°C thermal shock tests.
∞ Preferred die M22520/5-33 closure B, alternate die M22520/5-03 closure B.
Σ Preferred die M22520/5-35 closure B, alternate die M22520/5-03 closure A.
& Preferred die M22520/5-41 closure B, alternate die M22520/5-05 closure B, or -09 closure A.
~ Preferred die M22520/5-19 closure B, alternate die M22520/5-05 closure A or -11, -57, closure A.
ϕ Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

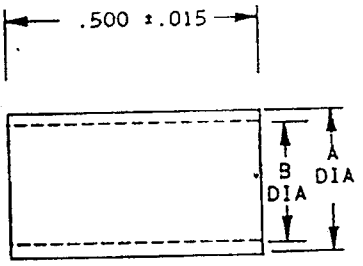
NOTE: Connectors mate with connectors of the same material only; i.e., M39012/57-3001 mates with M39012/55-3001, and M39012/57-4001 mates with M39012/55-4001.



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CENTER CONTACT




CRIMP FERRULE

| Dash number | Contact number 2/ | A ±.001 | Basic crimp tool 1/ | Crimp die or positioner | Crimp tensile minimum | Color band □ | Color band ○ |
|---|---|---|---|---|---|---|---|
| 3502<br>4502 | 57-10 | .041 | M22520/1-01 | Solder or M22520/1-15 | 4 pounds | Red | Maroon |

| Dash number | Ferrule number 2/ | A ±.003 | B ±.003 | Basic crimp tool 1/ | Crimp die or positioner M22520/5- |
|---|---|---|---|---|---|
| 3502<br>4502 | 57-50 | .250 | .220 | M22520/5-01 | -11, -05, -57 closure Closure A and -19 Closure B |

| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .003 | 0.08 |
| .005 | 0.13 |
| .010 | 0.25 |
| .015 | 0.38 |
| .024 | 0.61 |
| .035 | 0.89 |
| .038 | 0.97 |
| .041 | 1.04 |
| .050 | 1.27 |
| .062 | 1.57 |
| .133 | 3.38 |
| .153 | 3.89 |
| .193 | 4.90 |
| .220 | 5.59 |
| .250 | 6.35 |
| .326 | 8.28 |
| .500 | 12.70 |

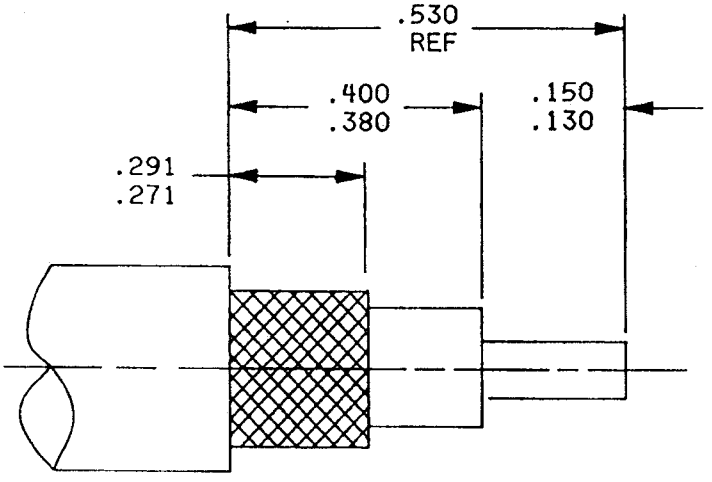
1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. .003 maximum break.
6. Color bands shall be positioned so that no coloring material enters the inspection hole.

FIGURE 3. Contact and ferrule dimensions for category D only.



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| Inches | mm |
|---|---|
| .130 | 3.30 |
| .150 | 3.81 |
| .271 | 6.88 |
| .291 | 7.39 |
| .380 | 9.65 |
| .400 | 10.16 |
| .530 | 13.46 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Cable stripping dimensions for field replaceable connectors.



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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 12,400 MHz.

Voltage rating: The voltage rating shall be in accordance with table II.

TABLE II. Voltage rating.

| Cables | Voltage max (at sea level) | Voltage max (70,000 ft) |
|---|---|---|
| | V rms | V rms |
| M17/93-RG178, M17/169-00001 | 170 | 45 |
| M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001 M17/152-00001 | 250 | 65 |
| M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001 | 335 | 85 |

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics:

Reference MIL-STD-348 and figure 5 dimensions.

Center contact (socket):

Oversize test pin: .0375+.0001.

Test pin finish: 16 microinches.

Insertion depth: .030/.045.

Number of insertions: 3.



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Insertion force test: Steel test pin diameter .0370+.0001.

Insertion depth: .050/.075.

Test pin finish: 16 microinches.

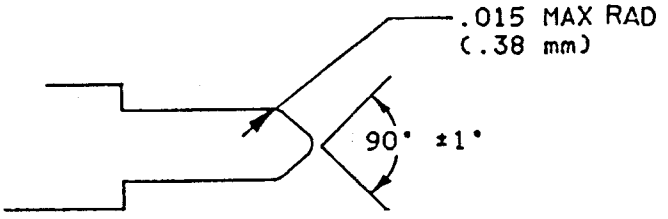
Insertion force: 2 pounds, maximum.

Withdrawal force test: Steel test pin diameter .0355 - .0001.

Insertion depth: .050/.075.

Withdrawal force: 1 ounce, minimum.

Test pin finish: 16 microinches.




NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 5. Test pin data.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B. 5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Radial torque: Not applicable.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 0.5 to 12.4 GHz, or approximately 80 percent of the cutoff frequency of the test cable, whichever is lower.

| Cables | VSWR |
|---|---|
| M17/93-RG178 | 1.20+0.025 (F) GHz |
| M17/054-RG122, M17/119-RG174, M17/113-RG316, M17/152-00001 | 1.15+0.02 (F) GHz |
| M17/028-RG058, M17/060-RG142, M17/084-RG223, M17/111-RG303, M17/128-RG400 | 1.15+0.01 (F) GHz |



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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025+.002 F (F in GHz).

Item 16: VSWR shall be less than 1.025+.002 F (F in GHz).

Second step of VSWR checkout procedure - VSWR shall be less than 1.080+.005 F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability:

Insertion and withdrawal force: 500 cycles, minimum at 12 cycles per minute, maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

| | Initial | After environment |
|---|---|---|
| Center contact: | 3.0 | 4.0 |
| Outer contact: | 2.0 | Not applicable |
| Braid to body: | 0.5 1/ | Not applicable |

Dielectric withstanding voltage at sea level:

Method 301 of MIL-STD-202.

| Cables | V rms |
|---|---|
| M17/93-RG178, M17/169-00001 | 500 |
| M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001 | 750 |
| M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001 | 1,000 |

Vibration, high frequency: Method 204 of MIL-STD-202, test condition D. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and V).

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

1/ Five milliohms are permissible on passivated steel bodied connectors.



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Corona level:

Altitude - 70,000 feet.

| Cables | Volts (min) |
|---|---|
| M17/93-RG178, M17/169-00001 | 125 |
| M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001 | 190 |
| M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001 | 250 |

RF high potential withstanding voltage:

Frequency: 5 to 7.5 MHz

Leakage current: Not applicable.

| Cables | Volts (min) |
|---|---|
| M17/93-RG178, M17/169-00001 | 335 |
| M17/054-RG122, M17/157-00001, M17/119-RG174, M17/173-00001, M17/113-RG316, M17/172-00001, M17/152-00001 | 500 |
| M17/028-RG058, M17/155-00001, M17/060-RG142, M17/158-00001, M17/084-RG223, M17/167-00001, M17/111-RG303, M17/170-00001, M17/128-RG400, M17/175-00001 | 670 |

Cable retention force: The cable retention force shall be in accordance with table III.

TABLE III. Cable retention force.

| Cable dielectric outer diameter | Pounds (min) | |
|---|---|---|
| | Single braid | Double braid |
| Inches (max) | | |
| .036 | 10 | N/A |
| .067 | 20 | N/A |
| .110 | 30 | N/A |
| .122 | 40 | 45 |

Coupling mechanism retention force: Not applicable.

RF leakage: -60 dB minimum tested at a frequency between 2 and 3 GHz.

RF insertion loss: dB max = .06 x $\sqrt{\text{freq GHz}}$. Test frequency at 6.0 GHz.

Part number: M39012/57 (dash number from table I or "B" number from table V).



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TABLE IV. Group qualification and retention testing.

| Group | Submission and qualification of any of the following connectors 1/ 2/ | Qualifies the following connectors |
|---|---|---|
| I | M39012/57- X009 | M39012/57- X006<br>M39012/57- X007<br>M39012/57- X008<br>M39012/57- X009<br>M39012/57- X010 |
| II | M39012/57B X015 | M39012/57B X011<br>M39012/57B X012<br>M39012/57B X013<br>M39012/57B X014<br>M39012/57B X015<br>M39012/57B X016<br>M39012/57B X017 |
| III | M39012/57B X022 | M39012/57B X018<br>M39012/57B X019<br>M39012/57B X020<br>M39012/57B X021<br>M39012/57B X022<br>M39012/57B X023<br>M39012/57B X024 |
| IV | M39012/57- X028 | M39012/57- X025<br>M39012/57- X026<br>M39012/57- X027<br>M39012/57- X028<br>M39012/57- X029 |
| V | M39012/57- X502 | M39012/57- X502 |

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same body material and finish only.

X denotes material.

NOTES:

1. For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified.

2. If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.



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TABLE V. Category B - Nonfield replaceable (special tools may be required).

Not for Air force or Navy use. For OEM use only

| M39012/57B | Applicable cable number = | Dimensions | Inches-millimeters maximum 1/ 2/ |
|---|---|---|---|
| 3011 3/ 4/<br>4011 3/ 4/ | M17/93-RG178*<br>M17/169-00001φ | | |
| 3012 3/ 4/<br>4012 3/ 4/ | M17/119-RG174<br>M17/173-00001φ<br>M17/113-RG316*<br>M17/172-00001φ | | |
| 3013 3/ 4/<br>4013 3/ 4/ | M17/54-RG122*<br>M17/157-00001φ | | |
| 3014 3/ 4/<br>[illegible] | M17/28-RG058*<br>[illegible] | A | 1.265 (32.13) |
| 3015 3/ 4/<br>4015 3/ 4/ | M17/60-RG142*a<br>M17/158-00001φ | | |
| 3016 3/ 4/<br>4016 3/ 4/ | M17/84-RG223*<br>M17/167-00001φ | | |
| 3017 3/ 4/<br>4017 3/ 4/ | M17/111-RG303*<br>M17/170-00001φ | | |
| 3018 5/<br>4018 5/ | M17/93-RG178*<br>M17/169-00001φ | | |
| 3019 5/<br>4019 5/ | M17/119-RG174<br>M17/173-00001φ<br>M17/113-RG316*<br>M17/172-00001φ | | |
| 3020 5/<br>4020 5/ | M17/54-RG122*<br>M17/157-00001φ | | |
| 3021 5/<br>4021 5/ | M17/28-RG058*<br>M17/155-00001φ | | |

See footnotes at end of table.



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TABLE V. Category B - Nonfield replaceable (special tools may be required) - Continued.

Not for Air force or Navy use. For OEM use only

| M39012/57B | Applicable cable number | Dimensions | Inches-millimeters maximum 1/ 2/ |
|---|---|---|---|
| 3022 5/<br>4022 5/ | M17/60-RG142*@<br>M17/158-00001ϕ | A | 1.265(32.13) |
| 3023 5/<br>4023 5/ | M17/84-RG223*<br>M17/167-00001ϕ | | |
| 3024 5/<br>4024 5/ | M17/111-RG303*<br>M17/170-00001ϕ | | |

1/ Millimeters are in parentheses.

2/ All corrosion resistant steel bodied connectors which are designed to be assembled to the cable outer conductor using solder shall be gold plated in accordance with MIL-G-45204, type II, class I.

3/ These connectors will be furnished with noncaptive center contacts by 1 January 1985.

4/ Inactive for new design.

5/ These connectors have captivated center contacts.

= The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in note Σ.

@ Cable to be used for the +200°C thermal shock tests.

ϕ Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

NOTE: Connectors mate with connectors of the same material only; Example: M39012/58-3001 mates with M39012/55-3001, and M39012/58-4001 mates with M39012/55-4001.



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TABLE VI. Maintenance replacements for category B.

| Category B number* | Category C dash number | Category A dash number | Category D dash number |
|---|---|---|---|
| BX011 | X025 | X006 | --- |
| BX012 | X026 | X007 | --- |
| BX013 | X027 | X008 | --- |
| BX014 | X029 | X009 | --- |
| BX015 | X028 | X009 | X502 |
| BX016 | X028 | X009 | --- |
| BX017 | X029 | X010 | --- |
| BX018 | X025 | X006 | --- |
| BX019 | X026 | X007 | --- |
| BX020 | X027 | X008 | --- |
| BX021 | X029 | X009 | --- |
| BX022 | X028 | X009 | X502 |
| BX023 | X028 | X009 | --- |
| BX024 | X029 | X010 | --- |

* Category B connectors are for original installation only. They will not be stocked or acquired by the Government.

X denotes material.

The material of the item shall be the same material as the item being replaced. Example: 55-3011 (corrosion resistant steel) replaces 55-3025.

TABLE VII. Cross reference of part numbers.

| Part number M39012/57B 1/ | Superseded part number M39012/57- |
|---|---|
| BX011 | 011 |
| BX012 | 012 |
| BX013 | 013 |
| BX014 | 014 |
| BX015 | 015 |
| BX016 | 016 |
| BX017 | 017 |
| BX018 | 018 |
| BX019 | 019 |
| BX020 | 020 |
| BX021 | 021 |
| BX022 | 022 |
| BX023 | 023 |
| BX024 | 024 |

1/ The 'B' part number is required marking for connectors manufactured after 3 April 1987. The connectors that are in stock or distribution that were previously qualified and marked with the old part number shall also be considered acceptable for Government use until stock is purged. (Applies to category 'B' part number change only; M39012/XXBXXXX).

The material of the item shall be the same material as the item being replaced. Example: 55B3011 (corrosion resistant steel) replaces 55-3025.



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Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - EA, MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV
Navy - AS, MC, OS, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3754-03)



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